Perfect Corp YouCam API Hackathon 2026

Mirror

The Decision Engine for Beauty Commerce

Problematic

The Core Insight:

- Consumers often struggle to know which beauty products, skincare routines, and clothing styles truly suit their unique appearance.
- Online shopping makes this harder because customers cannot easily visualize how products or clothes will look on them before purchasing.
- This leads to uncertainty, poor purchasing decisions, and potentially higher product returns for retailers.

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Three Distinct Market Pain Points

The Shopper

"Will this actually suit me?" A simple colour match does not answer whether a formula will cling to dry patches or shine on oily zones.

The Retailer (Traffic)

"I don't want to redirect customers." Retailers refuse to send high-intent shoppers to 3rd-party diagnostic platforms outside their store.

The Retailer (Tech)

"I have a catalogue, not an engineering team." Personalisation tools usually require custom data pipelines and manual product attribute tagging.

Our Sollution



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What we offer:

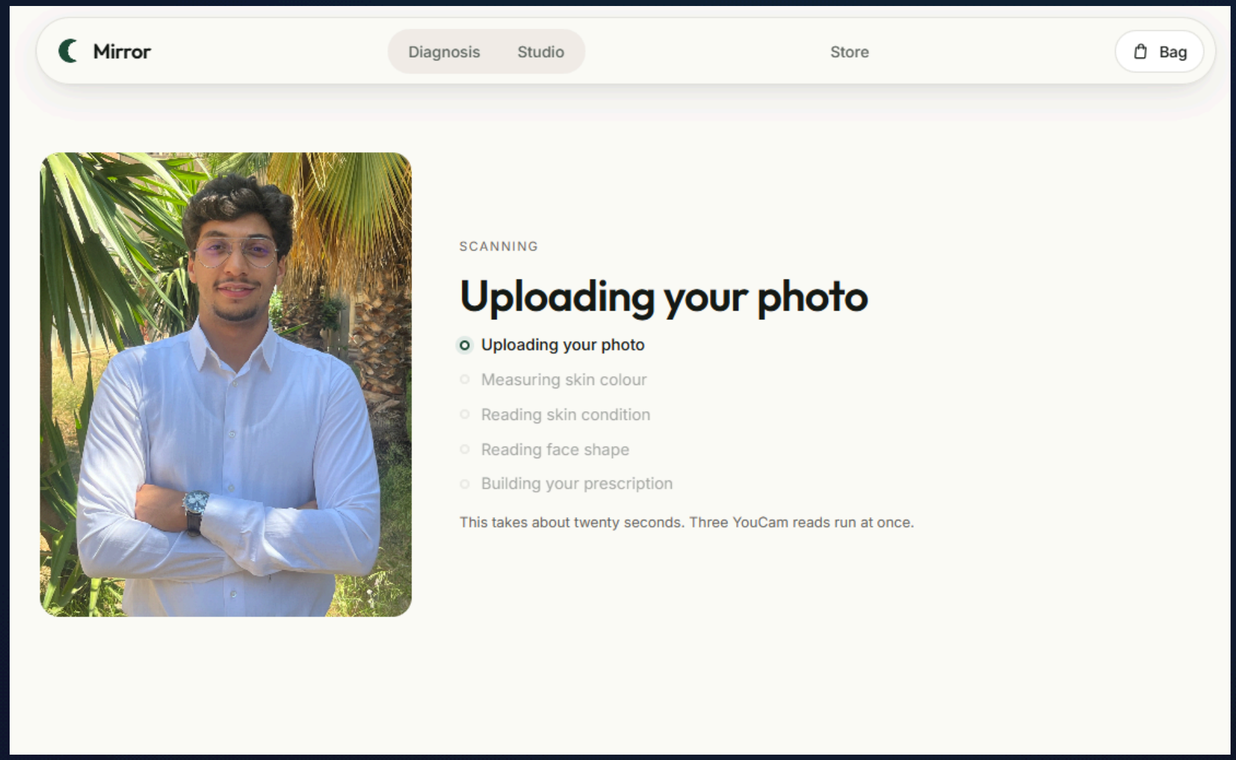
- Mirror reads one selfie and shops for you: it measures your skin's colour, which picks the shade, and your skin's condition, which picks the formula — the half every shade finder skips, and the reason two people who match the same foundation shade need opposite products.
- One scan sets a reading that six aisles reuse, each ranked by its own question — foundation must match your skin, clothes must flatter it, lipstick and blush must read as cosmetics on it — and every product names the measurement that chose it, down to the ΔE.
- The ranking is code you can audit rather than a model guessing: a language model only picks six from a shortlist of twelve the code already ranked, and when it's absent you get the same six, labelled as colour matches rather than advice.

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One Scan, 6 Aisles

Mirror extracts actionable biometric readings from a single photo to serve all retail categories without forcing re-scans.

- Single Upload: One photo produces three parallel YouCam readings.
- 7 Diagnostic Scores: Measures oiliness, moisture, redness, texture, pores, face shape, and tone.
- Objective Formulas: Directly maps metrics to finish, glow, coverage, and under-eye intensity.



User Flow: Scan to Checkout



The Demo: Same Shade, New Formula

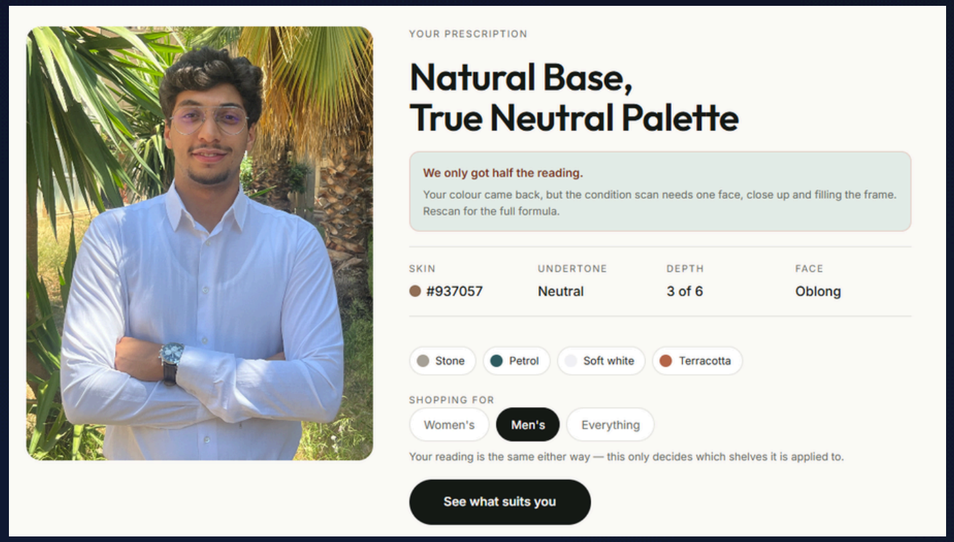
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User A: High Sebum Profile

Color Match: Shade #937057 (Medium Neutral)

Skin Condition: Oiliness Score: 28/100 (High Sebum), Pores: 34/100

Prescribed Product: Matte Finish, Oil-Control Complex, High Coverage

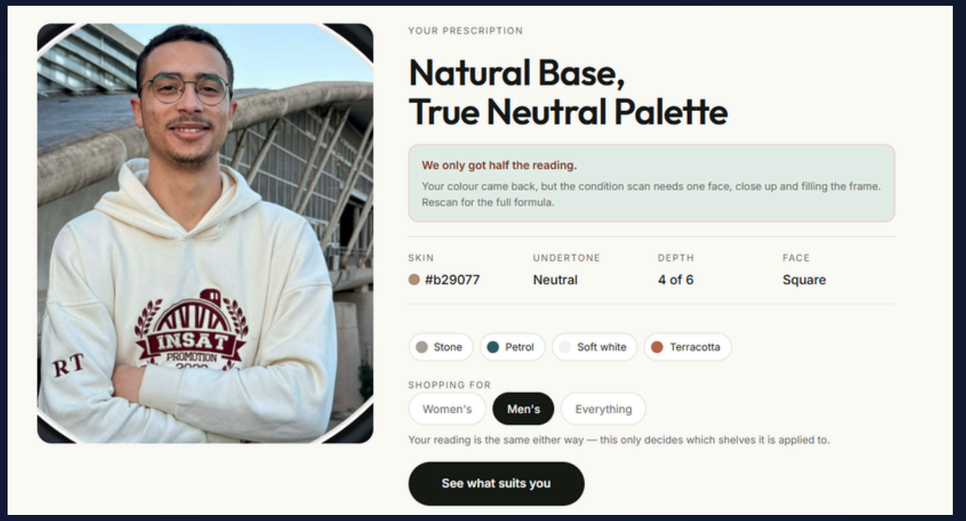


User B: Low Moisture Profile

Color Match: Shade #B29077 (Medium Neutral)

Skin Condition: Moisture Score: 82/100 (Dry/Dehydrated), Texture: 41/100

Prescribed Product: Dewy Finish, Hyaluronic Serum Base, Sheer Glow



Competitive Landscape & Matrix

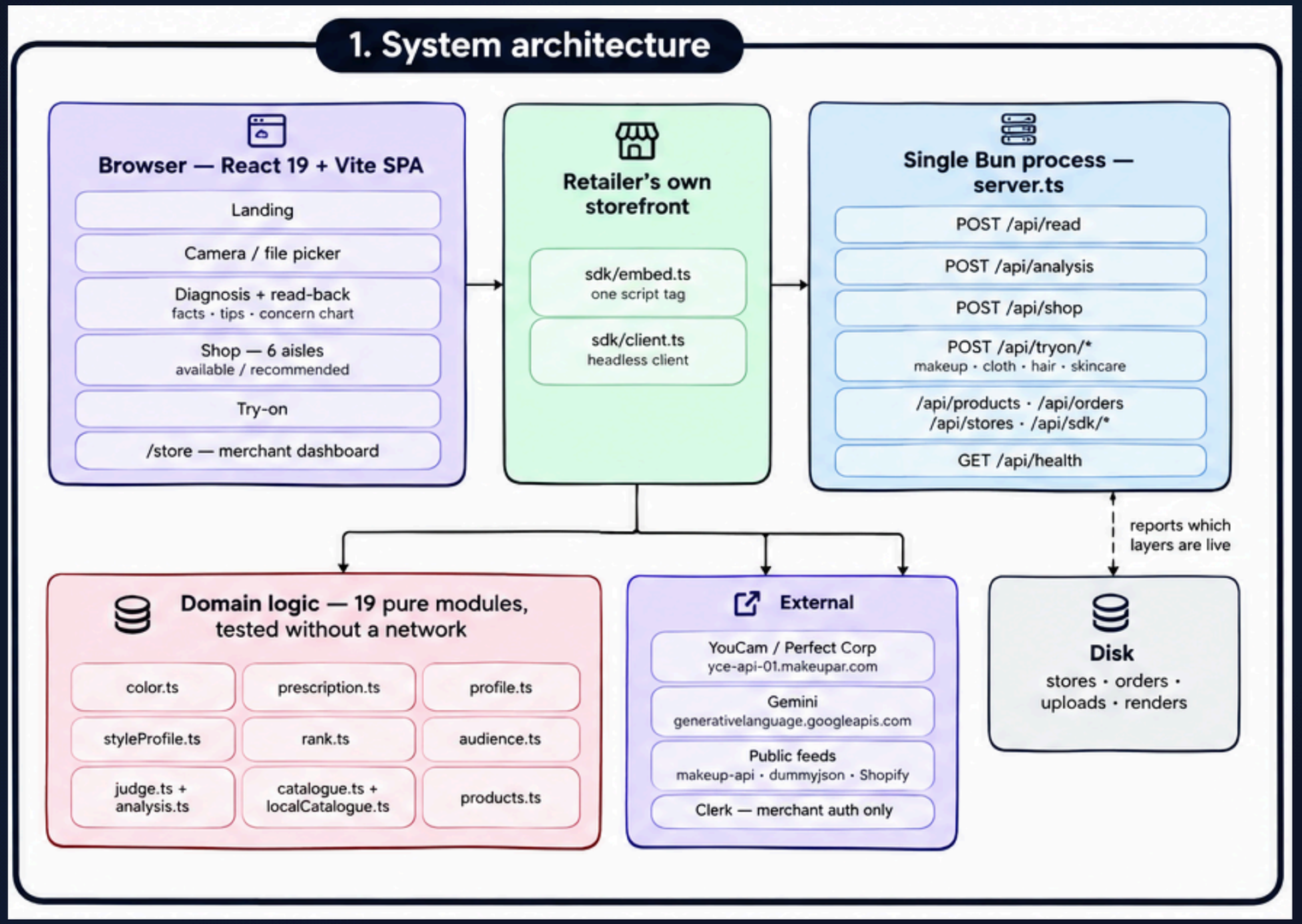
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Capability Axis	Shade Finders	AR Try-On	Skin Diagnostics	Mirror
Matches colour (CIELAB AE)	✓	✗	✗	✓
Diagnoses skin condition	✗	✗	✓	✓
Condition alters product formula	✗	✗	− Rarely	✓ Yes (Core)
Cites exact metric per item	✗	✗	✗	✓
Ranks merchant's own catalogue	✗	− Partly	✗	✓
Requires zero hand-tagging	✗	✗	✗	✓
Graceful fallback when AI down	N/A	N/A	N/A	✓ Yes (Labelled)

System Components

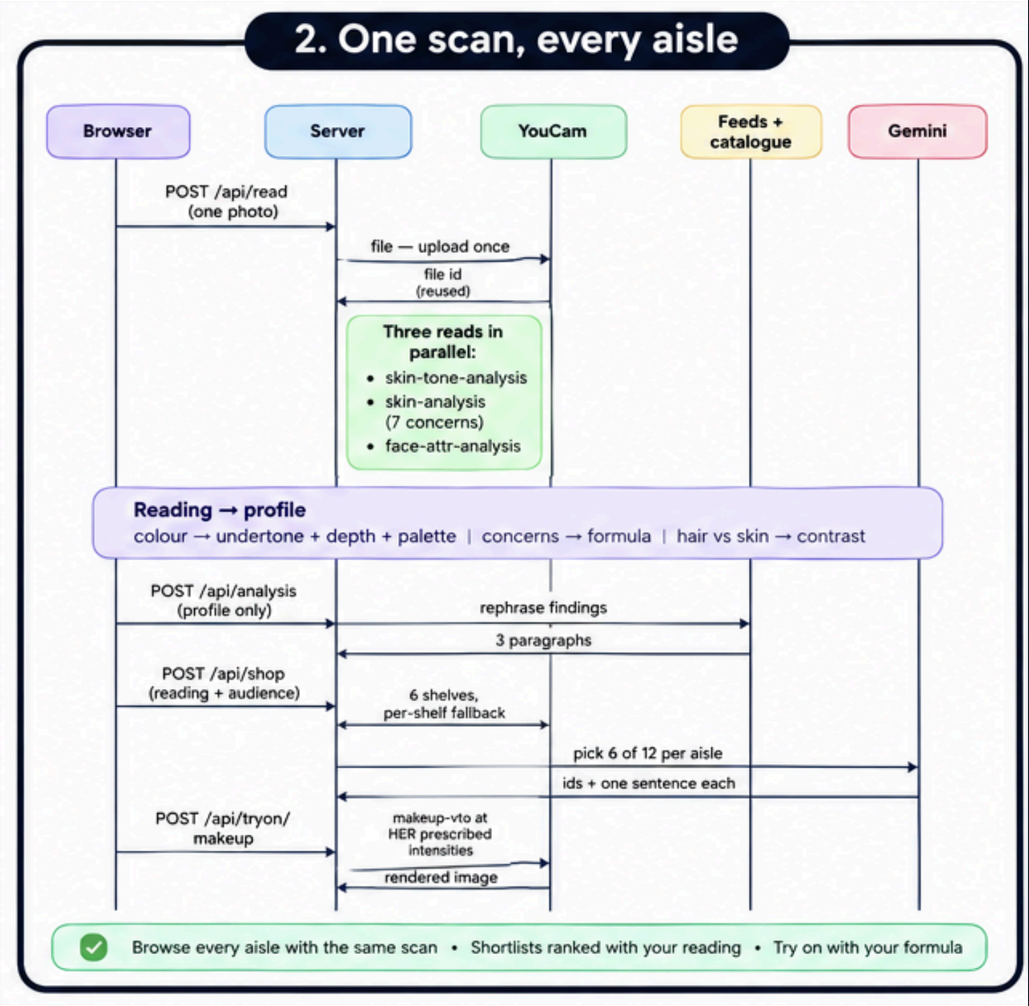
1. System Architecture

- The architecture connects the browser-based SPA, storefront, and single backend service.
- It integrates domain logic, external services, and storage to support the complete virtual try-on workflow.



2. Squences and working workflow

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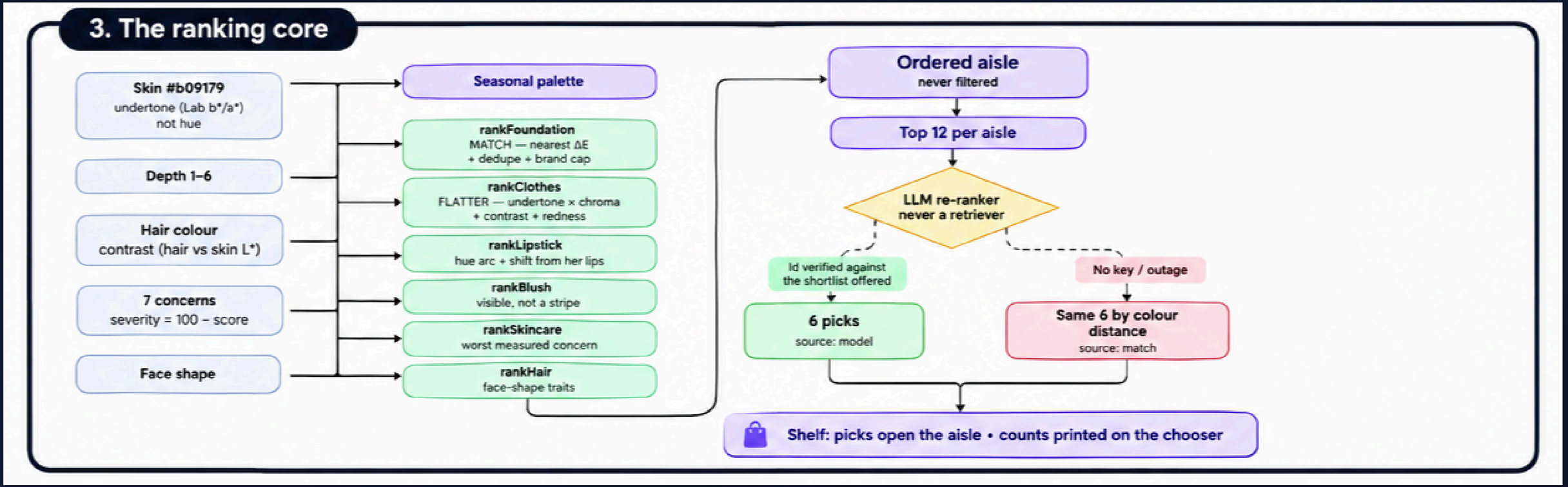


- **The workflow shows how the system processes a single scan and turns it into a complete personalized experience.**

3. The ranking score + recommendation system

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- The system ranks each product based on the user's profile and preferences.
- It then generates a personalized, ordered selection of the best recommendations.



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Our System

Algorithmic Ranking per Aisle

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Foundation (Match)

CIELAB ΔE color distance with near-duplicate filter ($\Delta E < 1.5$) and brand caps to prevent single-brand monopolization.

Clothes (Flutter)

Undertone weighting by color assertiveness + simultaneous contrast complement calculation to offset skin redness.

Lipstick / Blush (Enhance)

Gated to natural hue arcs; scored on bounded shift from measured lip tone (prevents costume looks).

Skincare & Hair

Skincare targeted at worst measured skin concern; Hair ranked directly against measured face shape metrics.

Deterministic LLM Re-Ranking

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Deterministic Pre-Filtering

Pure code ranks and narrows the catalogue down to a top 12 candidate list based strictly on color and formula metrics.

The LLM never acts as a database retriever, eliminating hallucinated products and inventory errors.

Guaranteed Fallback

The LLM selects 6 of 12 items and writes a personalized rationale for each.

If the LLM or network fails, the system serves the top 6 code-matched items labeled clearly as "Closest Match" rather than "Stylist Advice".

The Benefts

Zero-Tagging Merchant Onboarding

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1. Any Spreadsheet

Stores facing high return rates and endless color inquiries can simply upload product specs, letting shoppers try items on directly.

2. Server Measurement

All measurements are processed entirely on the server side, eliminating the need for high-performance hardware.

3. Single Script SDK

Retailers can easily integrate the try-on experience into their storefront using our custom-developed SDK.

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API Integration

Our Integration with YouCam API

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YouCam API Endpoint	Extracted Parameters & Metrics	Downstream Application
skin-tone-analysis	CIELAB L*, a*, b* color space vectors	Undertone, depth (1-6), seasonal color palette
skin-analysis	Spots, Wrinkles, Texture, Pores, Redness, Sebum, Moisture	Inverted to formula profile (matte vs dewy, coverage)
face-attr-analysis	Face shape geometry, hair color, eye color	Hair style recommendation, contrast ratio, wardrobe palette
virtual-try-on	Live image overlay rendering	Applies prescribed product intensity dynamically

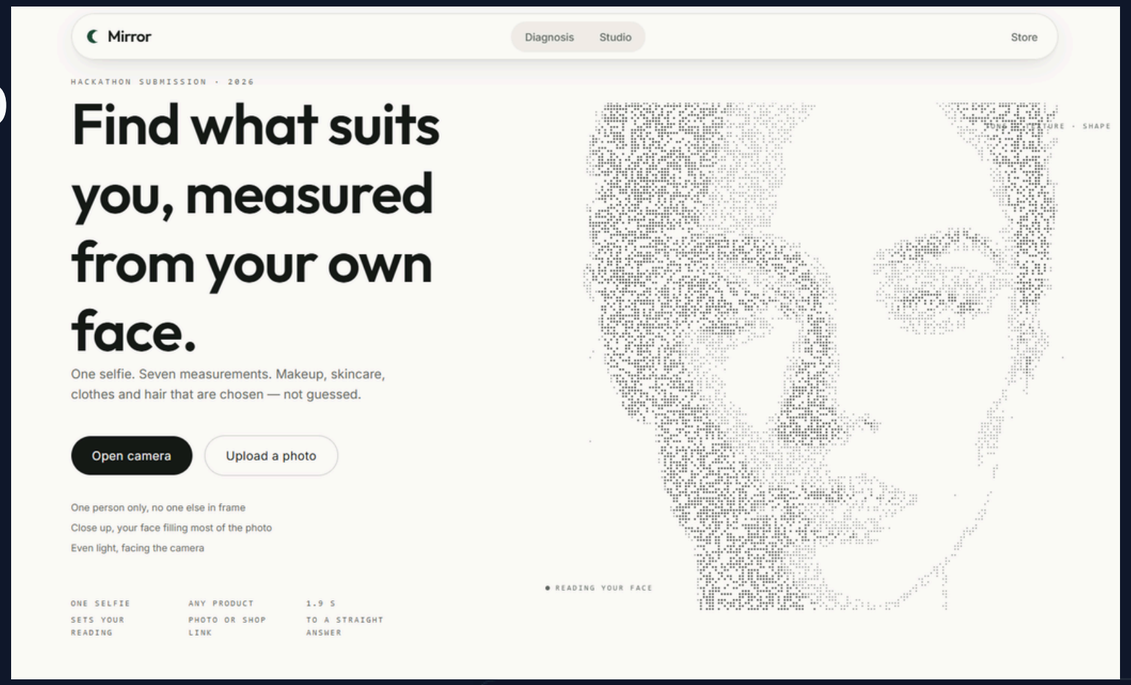
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UI/UX & Interfaces

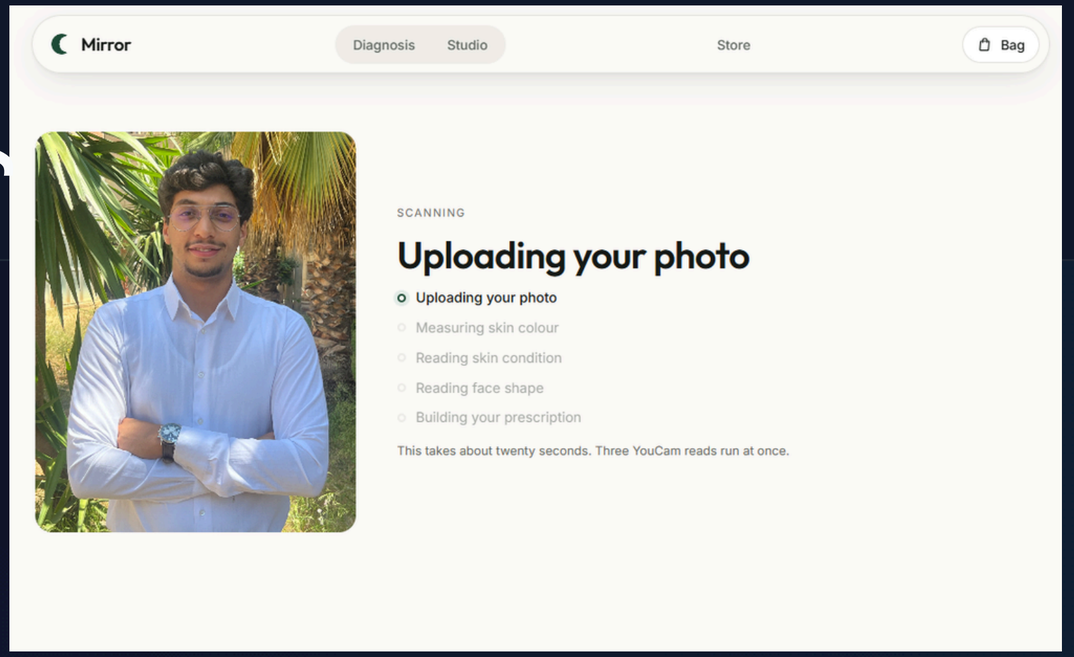
User interfaces

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Our Landing

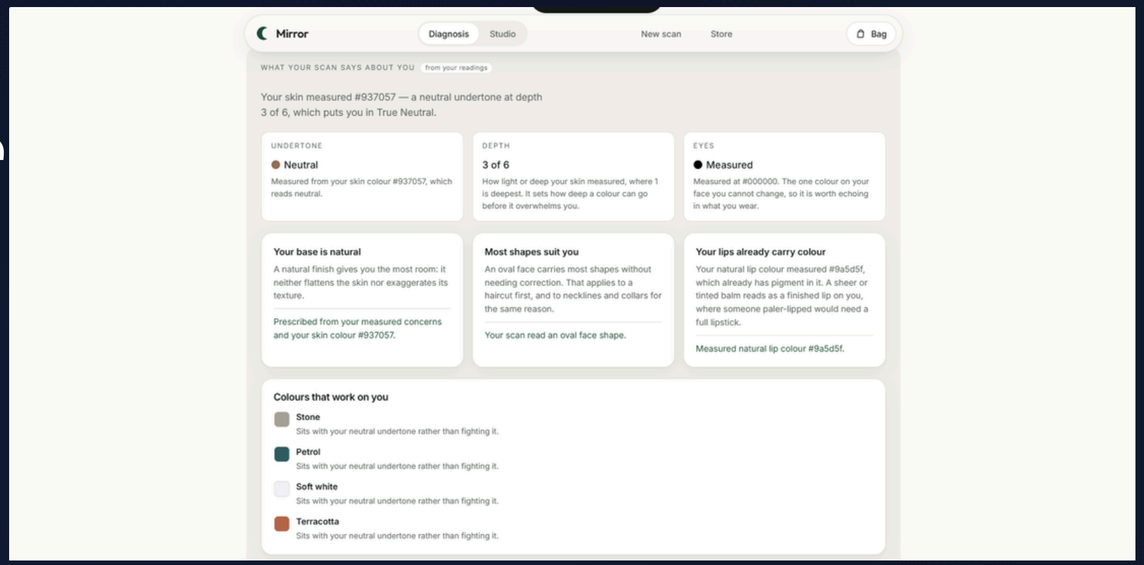


Profile analysis

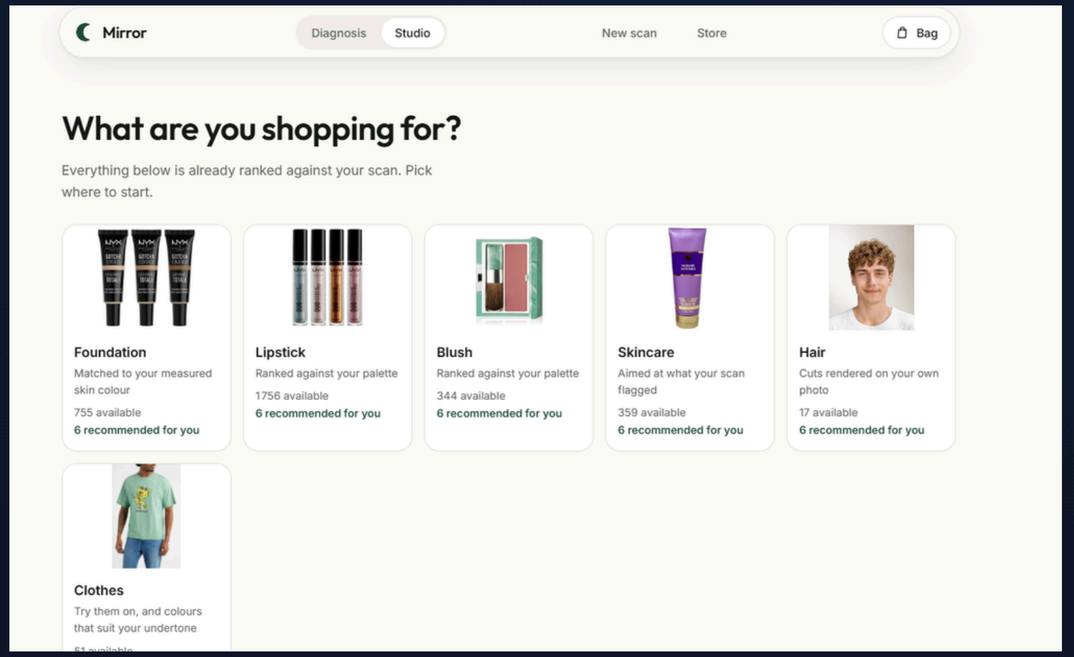


User interfaces

Final Analysis

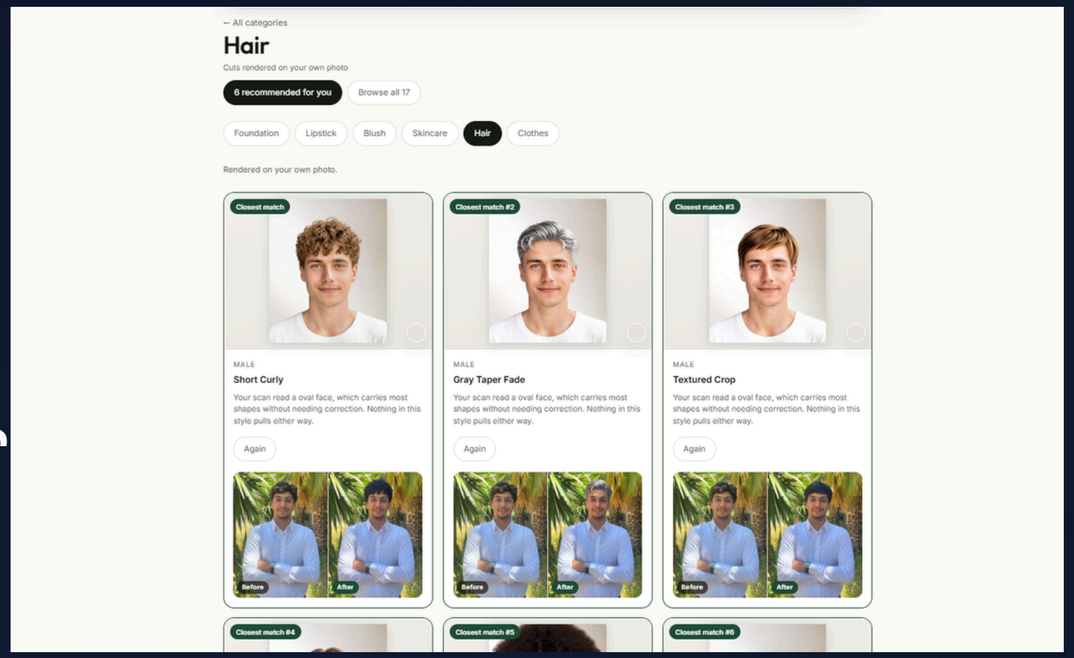


Different Categories

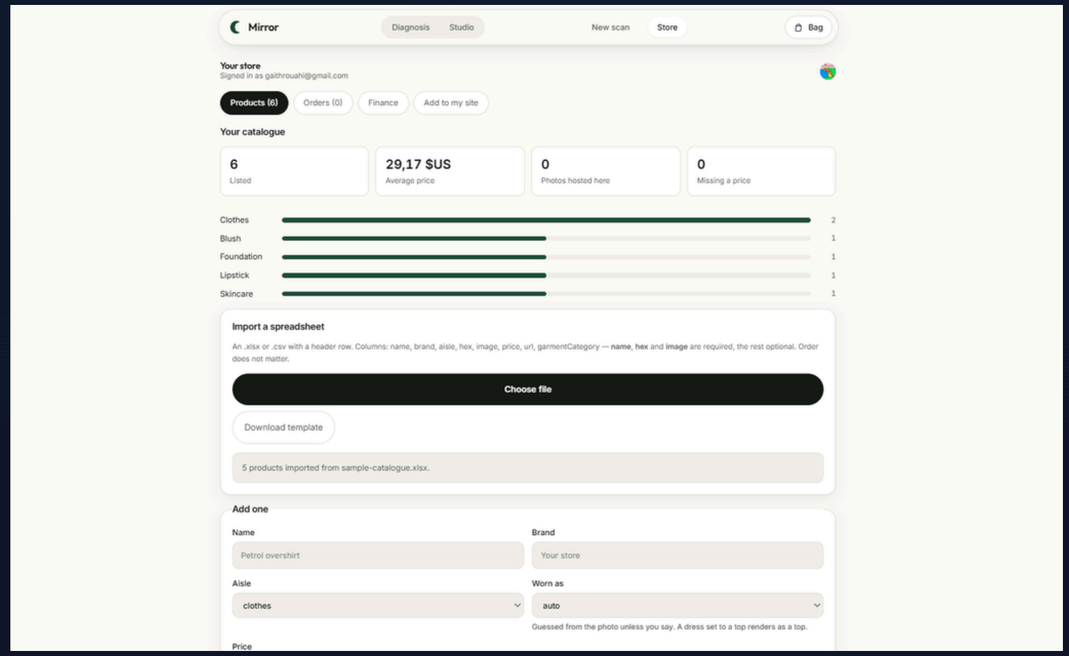


User interfaces

Try-on Core

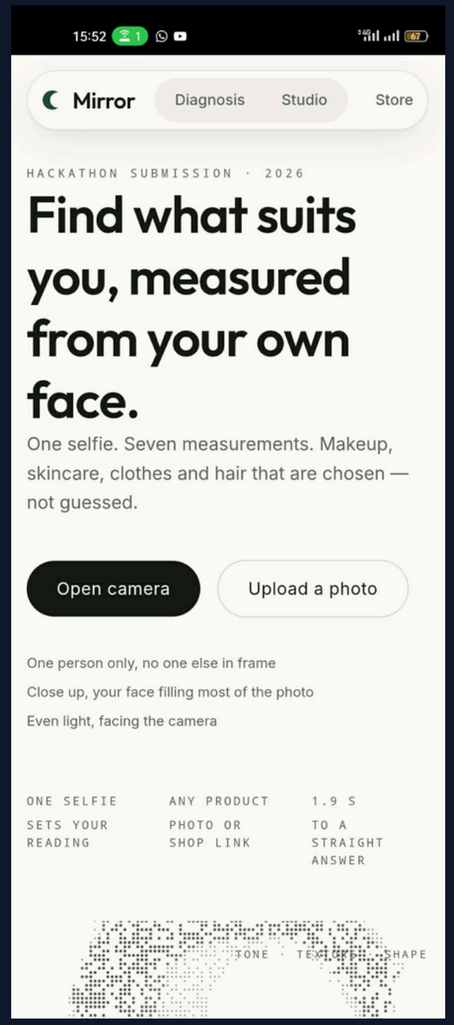


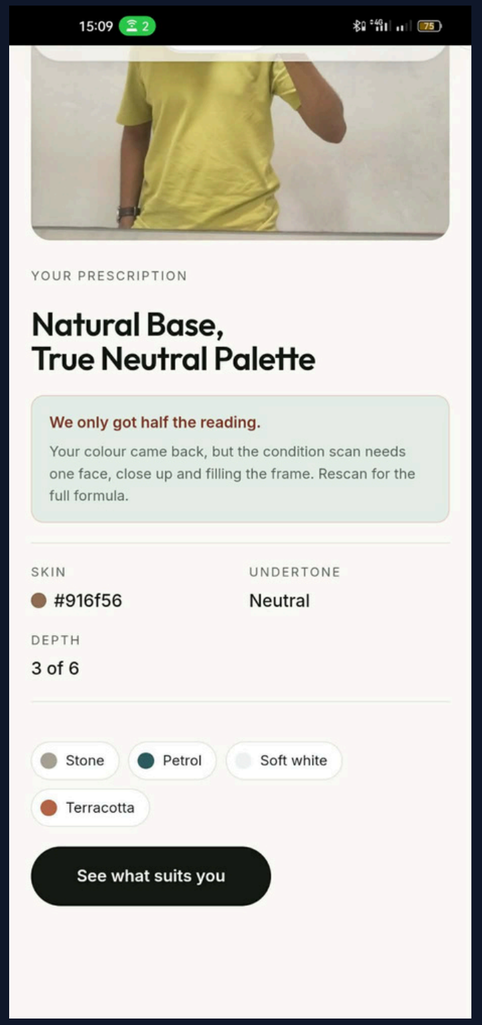
store

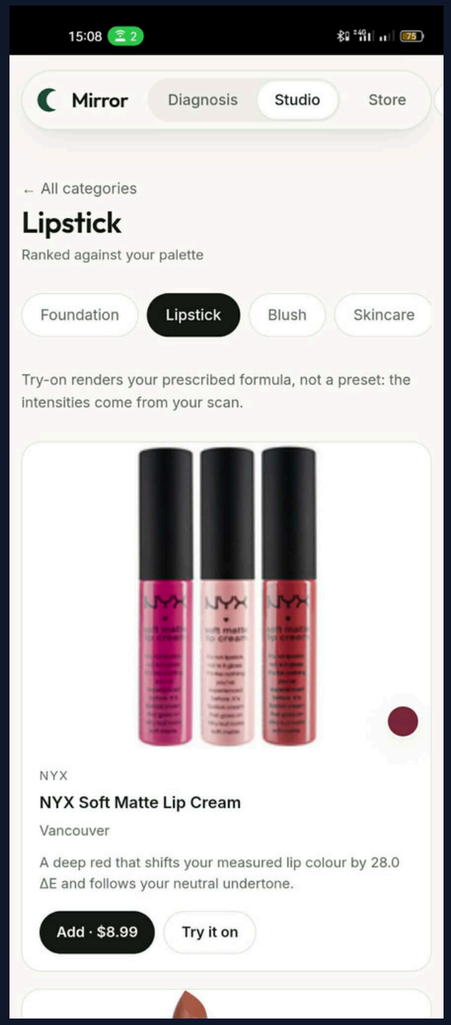


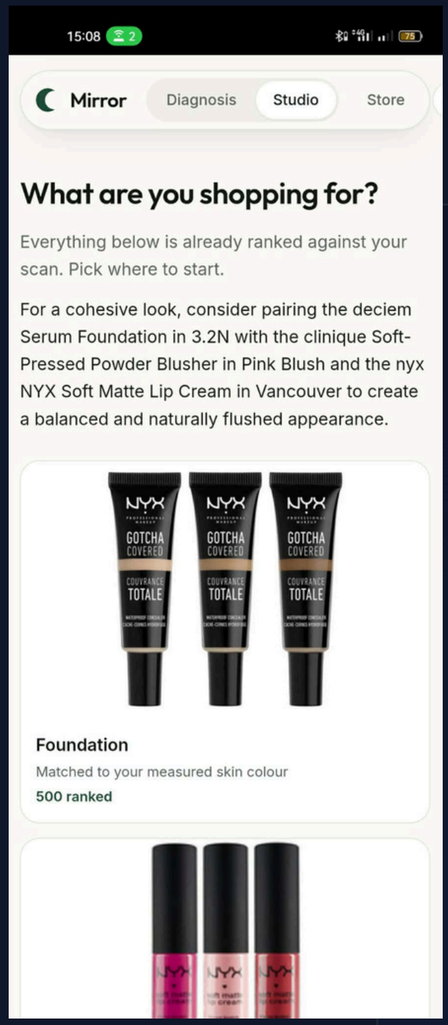
Mobile Application Views

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Thanks for your attention

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